

To find Machine Learning regression method using r2 value

1. MULTIPLE LINEAR REGRESSION= 0.9358680970046243

2. 2.SUPPORT VECTOR MACHINE (r2 value)

S.NO	HYPER PARAMETER	LINEAR	RBF (NON-LINEAR)	POLY	SIGMOID
1	C=10	-0.0396	-0.0568	-0.0536	-0.0547
2	C=100	0.1064	-0.0507	-0.0198	-0.0304
3	C=500	0.5928	-0.0243	0.1146	0.0705
4	C=1000	0.7802	0.0067	0.2661	0.1850
5	C=2000	0.8767	0.0675	0.4810	0.3970
6	C=3000	0.8956	0.1232	0.6370	0.5913

BEST COMBO :

C=3000,kernel=linear, r_score=0.8956

3.DECISION TREE

S.NO	CRITERION	MAX FEATURES	SPLITTER	R SCORE
1	absolute_error	None	best	0.9329
2	absolute_error	None	random	0.7275
3	absolute_error	sqrt	best	0.6032
4	absolute_error	sqrt	random	-1.1813
5	absolute_error	log2	best	0.6032
6	absolute_error	log2	random	- 1.1813
7	squared_error	None	best	0.9087

8	squared_error	None	random	0.6594
9	squared_error	sqrt	best	0.4218
10	squared_error	sqrt	random	0.7055
11	squared_error	log2	best	0.4218
12	squared_error	log2	random	0.7055
13	poisson	None	best	0.9134
14	poisson	None	random	0.6594
15	poisson	sqrt	best	0.2589
16	poisson	sqrt	random	0.7055
17	poisson	log2	best	0.2589
18	poisson	log2	random	0.7055
19	friedman_mse	None	best	0.8930
20	friedman_mse	None	random	0.6594
21	friedman_mse	sqrt	best	0.4218
22	friedman_mse	sqrt	random	0.7055
23	friedman_mse	log2	best	0.4218
24	friedman_mse	log2	random	0.7055

BEST COMBO:

Criterion: absolute_error, max_features=None, splitter=best r_score=0.9329